Classical Field Theory — Exercise Sheet 4

Formula, stability and weak derivatives

Hints and answers

Use the unitary Fourier transform $\widehat{f}(k) = (2\pi)^{-1/2}\int_{\mathbb{R}} e^{-ikx}f(x)\mathrm{d}x$ and $\|f\|_{H^1}^2 = \int_{\mathbb{R}}(1+k^2)|\widehat{f}(k)|^2\mathrm{d}k$. All wave equations on this sheet have one spatial dimension and zero mass. The solution class is that of Theorem 4.3.

Exercise 4.1 (Two initial data and two travelling packets). Let $f, g \in C_c^\infty(\mathbb{R};\mathbb{R})$ be initial displacement and velocity.

(a) Derive d'Alembert's formula from Lecture 4's Fourier multipliers. Use an oriented integral when $t < 0$. Verify the equation and both data, and justify identifying the formula with the energy-class solution.

(b) Let F, G be smooth real functions supported in $[-1/4, 1/4]$ and strictly positive on its interior. Take

$$f(x) = F(x+3) + 2G(x-2), \qquad g(x) = -F'(x+3) + 2G'(x-2).$$

Find the solution. For $t \geq 0$, find the open time intervals on which an observer at $x = 0$ sees each contribution. Where and when do the packet centres meet? Find the corresponding visibility interval at that position.

Hint. Write $\cos(tk)$ as two exponentials and $S_t(|k|) = \int_0^t \cos(sk)\mathrm{d}s$. Translation in physical space then turns the sine term into an integral of g.

Answer. $v = \frac{1}{2}[f(x+t)+f(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} g$. The packets are $F(x+3-t)+2G(x-2+t)$; at zero their intervals are $(11/4, 13/4)$ and $(7/4, 9/4)$. Their centres meet at $(t, x) = (5/2, -1/2)$; both are visible there for $t \in (9/4, 11/4)$.

Exercise 4.2 (Hadamard instability for an elliptic Cauchy problem). For each positive integer n, seek a smooth function $u_n(x, y)$, periodic in $x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, satisfying

$$\partial_y^2 u_n + \partial_x^2 u_n = 0, \qquad u_n(x, 0) = 0, \qquad \partial_y u_n(x, 0) = e^{-\sqrt{n}}\cos(nx).$$

Find such a solution by separation of variables. Prove that the initial velocity tends to zero in every fixed $C^r(\mathbb{T})$ seminorm, while $\|u_n(\cdot, y)\|_{L^2(\mathbb{T})} \to \infty$ for every fixed $y > 0$. What continuity claim does this disprove? Why does it not contradict Lecture 4?

Hint. The amplitude solves $a_n'' - n^2 a_n = 0$. Compare the logarithms of $n^r e^{-\sqrt{n}}$ and $e^{-\sqrt{n}}\sinh(ny)/n$ as $n \to \infty$.

Answer. $u_n = e^{-\sqrt{n}}\sinh(ny)\cos(nx)/n$ and $\|u_n(\cdot, y)\|_2 = \sqrt{\pi}e^{-\sqrt{n}}\sinh(ny)/n$. The elliptic Cauchy solution map from smooth data to the slice L^2 norm is discontinuous at zero; the equation has the opposite spatial sign to the wave equation.

Exercise 4.3 (A corner and its weak derivatives ⋆). Let $f(x) = e^{-|x|}$. Compute its unitary Fourier transform and its H^1 norm. Identify its first weak derivative by testing against every $\zeta \in C_c^\infty(\mathbb{R})$, and calculate its second distributional derivative. Decide whether f belongs to the space

$$H^2(\mathbb{R}) = \left\{ h \in L^2(\mathbb{R}) : \int_{\mathbb{R}} (1+k^2)^2 |\widehat{h}(k)|^2 \mathrm{d}k < \infty \right\}.$$

Distinguish the value assigned to a weak derivative at the single point zero from the existence of a classical derivative there.

Hint. Split each integral at zero. The endpoint terms cancel for the first derivative and add for the derivative of the first derivative.

Answer. $\widehat{f} = \sqrt{2/\pi}/(1+k^2)$, $\|f\|_{H^1}^2 = 2$, $f' = -\operatorname{sgn}(x) e^{-|x|}$ almost everywhere, and $f'' = f - 2\delta_0$. Thus $f \in H^1 \setminus H^2$ and has no classical derivative at zero.

Exercise 4.4 (The sharp massless sine bound). Choose a real even $\rho \in C_c^\infty((-1,1))$ with $\|\rho\|_2 = 1$. For $0 < \varepsilon \le 1$, let g_ε satisfy $\widehat{g}_\varepsilon(k) = \varepsilon^{-1/2} \rho(k/\varepsilon)$, and let v_ε have initial data $(0, g_\varepsilon)$.

(a) Prove that g_ε is real and normalized in L^2, and that for every $T > 0$,

$$\sup_{|t| \le T} \|v_\varepsilon(t) - t g_\varepsilon\|_2 \le \frac{T^3 \varepsilon^2}{6}.$$

(b) For fixed real t, determine the operator norm on real $L^2(\mathbb{R})$ of the Fourier multiplier $S_t(|k|)$, where $S_t(0) = t$. Explain why conservation of massless energy cannot give a bound for $\|v(t)\|_2$ uniform in both time and all initial data with energy $1/2$.

Hint. Integrate the estimate $|1 - \cos s| \le s^2/2$ to control $\sin z - z$. The shrinking Fourier support makes the sine multiplier approach its zero-frequency value uniformly on each bounded time interval.

Answer. $\|g_\varepsilon\|_2 = 1$ and the stated error bound holds. The multiplier norm is exactly $|t|$, including zero at $t = 0$. All these data have massless energy $1/2$, but their displacement norms approach $|t|$ for every fixed t as $\varepsilon \to 0$.